TKT-BCS
Operation Manual

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1-General Description

Thank you for choosing TKT products. In case of any doubt please contact the nearest.

TKT's authorized net of services or contact the technical assistance department.

The equipments are simple to operate and have great precision of control. The systems with compact dimensions, guarantee a high cooling capacity with low noise level.

This manual was developed with the purpose of presenting important functioning aspects,operation and maintenance, to obtain the best performance out of the air conditioning system.

To ensure a long useful and problem free life to the equipment it is essential that the operation and maintenance instructions described in this manual are followed and performed periodically.

TKT keeps a net of authorized services with tools, equipments and a team of professionals trained to perform any type of maintenance within the quality standard.

Attention:

In case any problem in the cooling circuit occurs, the repair must be carried out by an TKT authorized service station, or an expert.

Revision Status:

TKT is striving to improve the quality of products hence information in this manual may therefore changed at any time without notice.

Name Plate:

Each unit can be identified by its name-plate for further correspondence.

Refrigerant:

The refrigerant used in this system is Environmental friendly R134a.The refrigerant contained in the system of your unit can cause frostbite, serve burns, or blindness when in direct contact with skin or eyes hence proper care should be taken to avoid injuries.

The refrigerant capacities can vary depending upon the length of the refrigerant hose. The length of the hoses varies depending upon chassis length, location of system & the compressor location.

2-Commissioning of Battery Cooling System

The continued proper functioning of the system depends on through professional standards of work right from the outset. This entails proper & careful handling of tools like vacuum pumps, charging lines etc. will provide many years of reliable system operation.

Vacuuming:

It should has following purpose.

To remove air

To remove moisture.

To remove extraneous gases.

The vacuum pump should be powerful enough to maintain an absolute level of 0.3 mbar is required. A high level vacuum required in order being certain air & extraneous gas is drawn off all the time. Dropping the pressure to 0.3 to 1 mbar will ensure that boiling point of water to about 0 ℃， causing any moisture in the system to condensate out & the water vapour to be drawn off with the air & extraneous gases.

Operating procedure:

Open the sheet metal cover above the BTMS. Open the high and low pressure filling valve caps of BTMS, connect the high and low pressure pipelines of the manifold gauge, and connect the intermediate pipeline to the vacuum pump. Turn on the vacuum pump, open the high and low pressure filling valves, start vacuuming, and after 20 minutes, close the valve to turn off the vacuum pump. Observe that there is no change in the pressure gauge.

Charging the Refrigerant:

If the battery cooling system is not running, the system can only be filled with liquid refrigerant.

The charge qty should be added by weight according to rating plate.

Operating procedure:

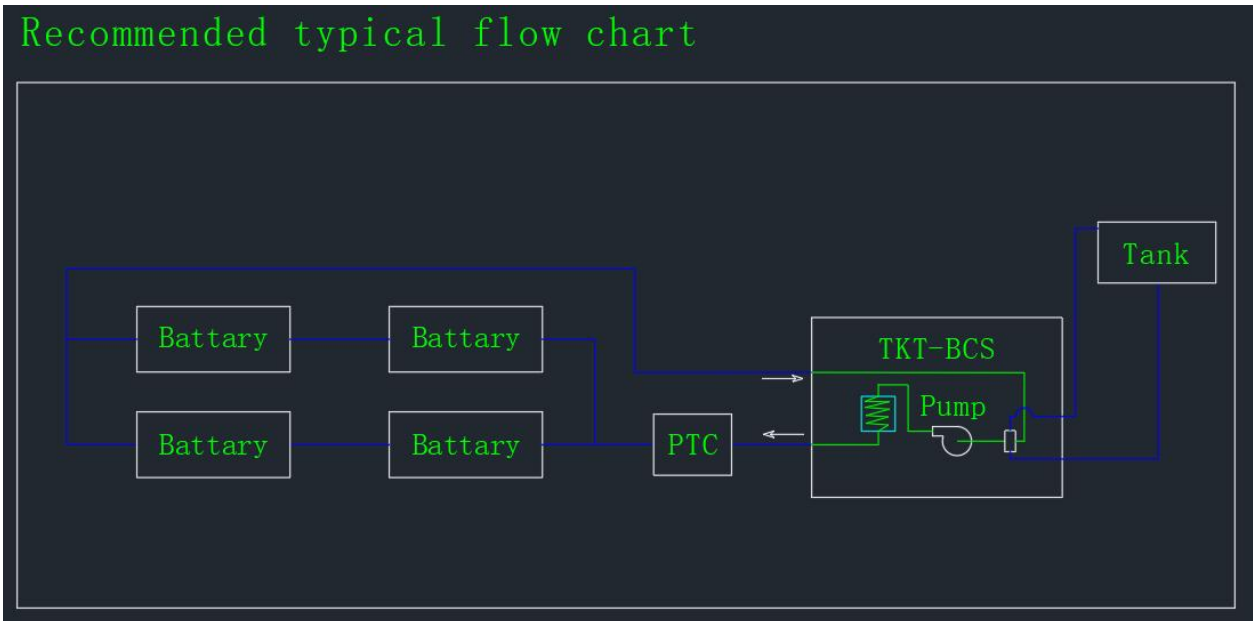
Disconnect the middle connecting pipe of the manifold gauge from the vacuum pump, connect the R134a refrigerant steel cylinder, invert the refrigerant steel cylinder on the scale, open the high and low pressure valves to start filling, and when filling for a period of time (when the weight of the scale no longer changes), close the high and low pressure filling valves. Open the BTMS system and let the compressor start working. Open the low-pressure filling valve to continue filling until the refrigerant filling amount reaches the weight required by the nameplate.

After filling the refrigerant, use soapy water to check for refrigerant leakage at the high and low pressure interfaces.

Charging the Water and exhaust:

1. Install the BTMS on the vehicle. The inlet and outlet pipes of the BTMS are

respectively connected to the inlet and outlet pipes of the battery. The makeup pipe and overflow pipe are connected to the expansion tank to ensure that the water system pipes are correctly connected. (Refer to the typical flowchart recommended in the drawing)



2. Connect the high and low voltage harnesses of BTMS to the vehicle, and then connect the BTMS debugging panel.

3. Add antifreeze to the expansion tank to the maximum scale line and wait for 2min. When the liquid level in the expansion tank no longer changes, the water pump is forced to start through the commissioning panel to allow the water circuit system to start circulating exhaust. As the liquid level in the expansion tank continues to drop, continue to add antifreeze to make it not lower than the minimum scale line. It takes about 15min, and the liquid level in the expansion tank basically remains unchanged, The gas in the waterway system has been evacuated. Then continue to add antifreeze to the maximum mark.

Turn on battery cooling:

By adjusting the panel to set the target temperature (lower than the actual temperature of the BTMS outlet), the BMS sends a cooling command to the BTMS. The water pump and condensing fan start first, and then the compressor starts running. The system starts cooling until the BTMS outlet temperature reaches the target temperature, and the system stops working.There are no issues with system debugging.

Turn on battery heating:

By adjusting the panel to set the target temperature (higher than the actual temperature of the BTMS outlet), the BMS sends a heating command to the BTMS. The water pump start first, and then the PTC starts working. The system starts heating until the BTMS outlet temperature reaches the target temperature, and the system stops working.There are no issues with system debugging.

3-Maintenance Schedule:

Performed monthly or every 250 hr

Check the condenser & clean it if necessary. Straighten up any fins / sheet bents. If coil is dirty it recommended clean of against the exhaust direction using compressed air or using a non-aggressive copper & aluminum cleaner.

Inspect all the hoses about fractures & loose connections & prevent contact with sharp edges or source of heat.

Check fixing screws & compressor fittings. If necessary, tighten or replace defective /worn parts.

Maintenance work to be performed annually or every 1000 hrs

Specialist in air conditioning must carry out these maintenance activities.

Work to be performed

Check cooling system for leaks.

Check the Refrigerant hose routing & fittings for anchorage, leaks &correct positioning.

Check high pressure & low-pressure switches are working properly according to specified limits.

Check the working condition of condenser.

Check the Relays & fuses. Check the compressor oil, if contaminated, it is recommended to replace.

Check the condition of electrical systems.

4-Service Instructions:

The efficient performance of the unit is dependent upon a proper maintenance & function test.

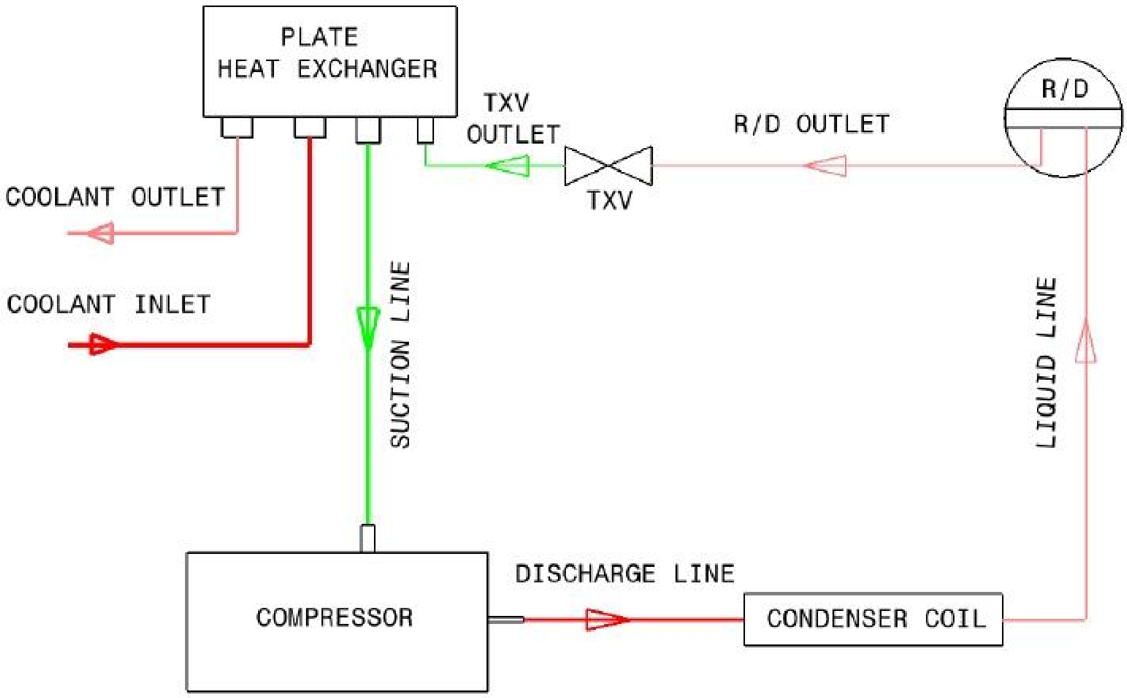
Visual inspection & function test by the driver will avoid expensive repairs, little faults are often responsible for big repairs hence working reliability of the BTMS depends upon a regular maintenance.

To keep the unit are operated in extreme dusty areas, maintenance intervals have to be adjusted to the needs.

If the BTMS are not used for long term for Example in the wintertime, it is recommended to operate at least once a month for minimum of 20 minutes. This prevents gaskets & bearings at the compressor will lubricated by it's oil pump.

The refrigeration system is a closed circuit, that why total cleanliness must be the aim of service engineer. Once a contaminant enters the system it will remain there, only intervention by a service engineer can remove it. Contaminate usually react slowly. A system may start up initially & run perfectly but a few months later it will be found to be badly damaged, perhaps beyond repairs.

5-Refrigeration & Coolant Flow Diagram:



6-TKT warranty terms:

TKT warranties its products for the period of one year, counted from the equipment installation date, which is stated on the cover of the warranty certificate.

No claim will be accepted if the vehicle continues to be used after the evidence of defect, even if there is lack of parts, delays in transportation or any other incident.

1 – The warranty will be valid for the period above specified, counted from the equipment installation date made by the first buyer/consumer, even though the property has been transferred.

2 – During the stipulated period, the warranty covers totally the workmanship and spare parts when repairing defects duly identified as being: equipment manufacturing; primary failure of material and defects of components used on its manufacturing.

3 – Only a technician from the TKT authorized net of services is qualified to repair the defects covered by the warranty.

4 – The repair or replacement of defective parts, performed by an authorized service station, will not incur in any debit regarding parts and workmanship used.

5 – The warranty approval is conditioned to the technical analysis of the defect presented on the components and operational conditions to which the equipment was submitted.

6 – The warranty of the components used on the assembly of the TKT equipment, which have their own net of technical assistance, will be obtained from its own net, by presenting the TKT warranty certificate.

7 – The warranty loses its validity:

a) If the installation or use of the product is not in accordance with the TKT technical recommendations.

b) If the product suffers any damage caused by accident, nature agents, misuse, or yet alterations and repairs performed by any other than the manufacturer's authorized technicians.

c) If the warranty certificate and/or the serial number of the product is adulterated, overwritten or damaged.

d) If defects or unsatisfactory performance are caused by the use of not original spare parts and in disagree with the technical specifications from SPHEROS.

8– The warranty not cover:

a) Transportation expenses to the repair facility. In case the consumer decides to be attended in the same place the product operates, the collection or not of the visit tax will be the criterion of the Authorized Service.

b) The attendance to the consumer, free or paid, in cities that do not have Authorized Services.Being thus the expenses with displacement are of total responsibility of the owner.

c) Failure of proper preventative maintenance, according to the described in this manual, on the preventative maintenance item.

d) Revisions, external adjustments and cleaning, therefore these information are listed on the owner’s guide.

e) Consumables Parts are not covered under warranty which are.Are considered, belts, filters in general, lubricant oil, relays and fuses.

f) TKT will not assume any losses caused A/c for non functioning equipment.